

IaaS To Manage ENTERPRISE COMPUTING RESOURCES

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IaaS provides cloud-based computing infrastructure that can help enterprises optimise their IT operations and resource management, improve troubleshooting and reduce operational costs significantly

Infrastructure as a service (IaaS) is a form of cloud computing infrastructure that allows virtual computing resources to be managed over the Internet. Cloud computing services have three categories: IaaS, platform as a service (PaaS) and software as a service (SaaS).

IaaS is a self-service model that delivers computer infrastructure to support enterprise operations for accessing, monitoring and managing remote data centres. It handles software, hardware, storage, computation, network components, networking services (like firewall), servers and data centres, and provides all computing resources to support the delivery of applications.

An IaaS provider is responsible for operating and maintaining equipment, databases, messaging queues and other policy-based

services above virtualisation. Without any major capital investment in additional hardware, organisations can use IaaS to fulfill additional computing capacity on-demand. Using IaaS, they can manage data, operating systems (OSes), runtimes, middleware and applications. They can also update these to newly-released versions.

Within a service provider's infrastructure, organisations use their own platforms and applications. An enterprise cloud is a hosted computing environment that delivers IaaS, PaaS and SaaS to business users via a network. Enterprise IT provides the foundational building blocks for businesses across various verticals, including access, data, interconnectivity, interoperability solutions and infrastructure.

Enterprise cloud computing (ECC) enables organisations to deliver computing services in a controlled and secure manner, using a network firewall. Government agencies and healthcare organisations that store, manage or process sensitive data use ECC services, instead of public cloud computing services.

ECC can be implemented in one of two ways: as a private cloud hosted on the premises of an organisation, or as a private cloud hosted externally by a third-party provider. Organisations that use externally-hosted ECC services enjoy the benefits of lower IT infrastructure, maintenance and IT operational expenses due to a decreased need for internal IT support staff.

Need for IaaS

Phanikishore Burre, head of delivery - cloud, infrastructure and security services, CSS Corp., says "Businesses are constantly looking for ways to streamline operations, accelerate business processes, reduce costs and increase service agility. In this context, they need to be intelligent, adaptive and agile when it comes to handling newer expectations. Despite significant adoption and investments in newer technologies, enter-

Fig. 1: Cloud computing layers

